

Audio Transcription using Whisper AI

 Converting Speech to Text with OpenAI's Whisper Model

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Cyfuture assignment



Made with **GAMMA**



Problem Statement & Objective

Problem Statement

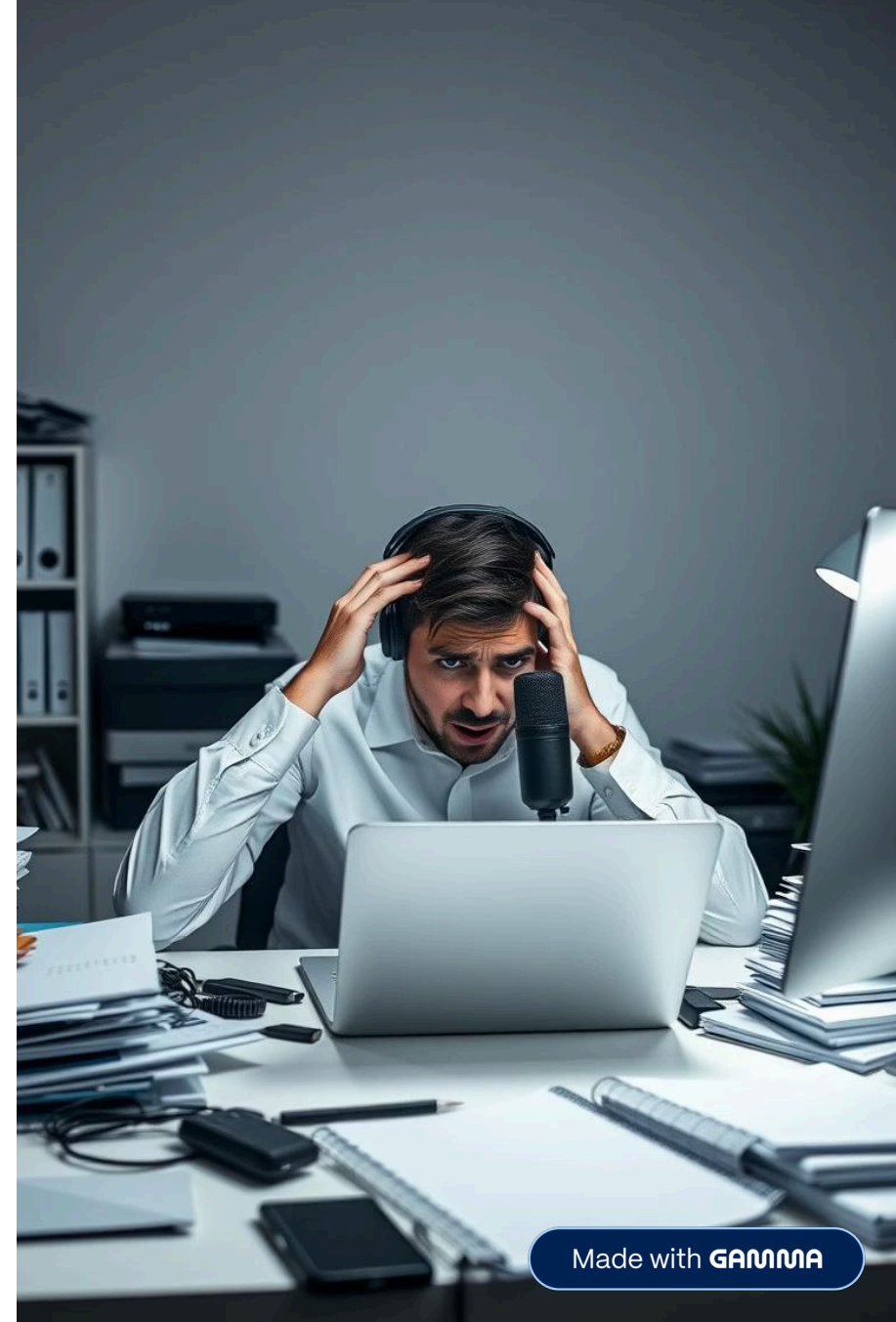
Manual transcription is slow and inaccurate.

Objective

Automate speech-to-text with OpenAI's Whisper model.

Goal

Fast and accurate transcription of MP3 audio files.



Method

Model Selection

OpenAI's Whisper models: Tiny, Base, Small, Medium, Large.

Input

MP3 audio files as source data.

Output

Transcribed text formats: .txt, .srt, .vtt.



Implementation Details

1

Python Environment

Use libraries like openai-whisper and pydub.

2

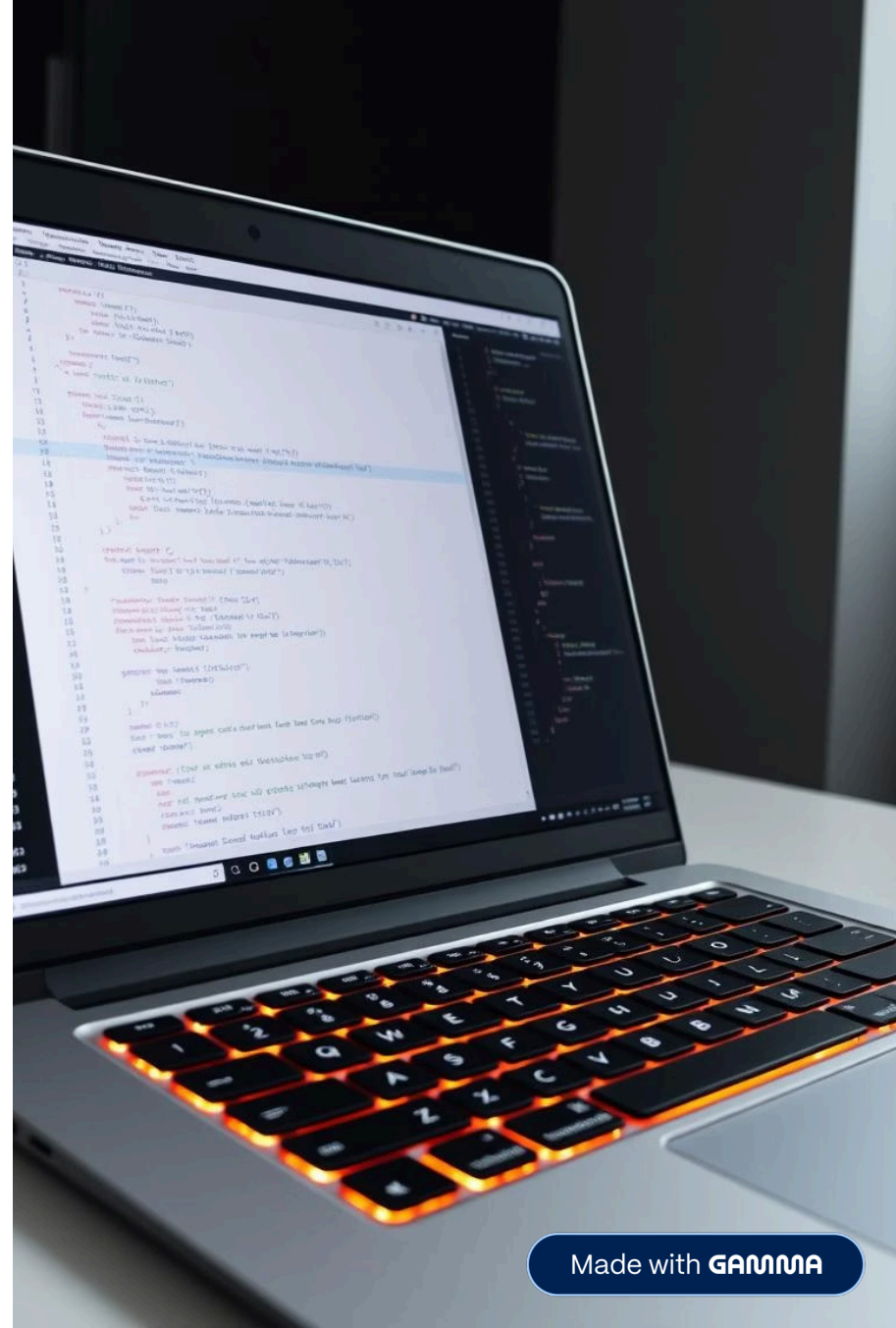
Code Workflow

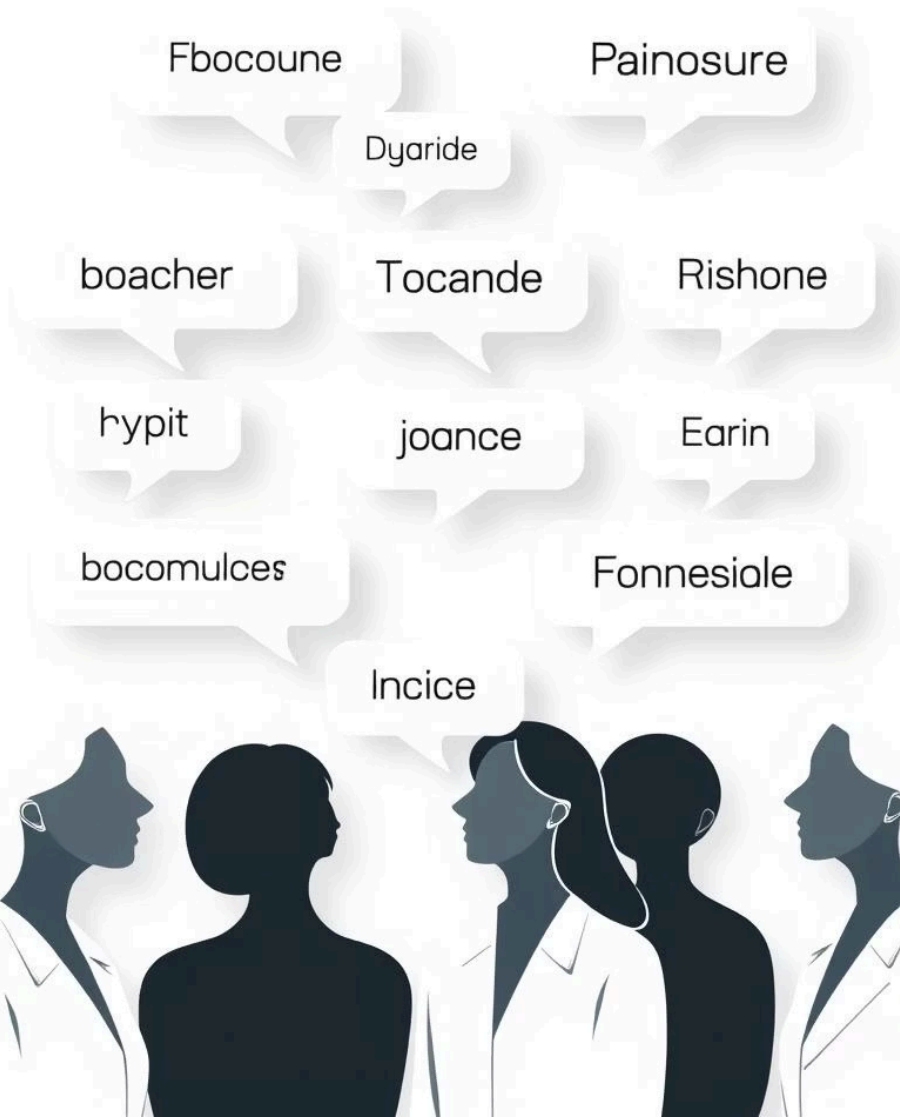
Load model, transcribe audio, save output files.

3

GPU Acceleration

Set up CUDA for faster transcription processing.





Advantages of Whisper AI

Multilingual Support

Transcribes in multiple languages effortlessly.

Noise & Accent Handling

Works well despite background noise and various accents.

Open-source & Flexible

Customizable for unique transcription needs.



Conclusion and Future Work

Revolutionizing Transcription

Whisper AI offers fast, accurate audio-to-text conversion.

Thank you and open for Q&A.

Future Enhancements

Real-time transcription and speaker diarization integration.

Next Steps

Expand AI tool integration and improve usability.